MILITARY SPECIFICATION

EARPHONE ELEMENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a dynamic earphone element having an impedance of 20 ohms with an extended frequency and a flat response (see 6.1). The element is part of the earcup assembly of the M83511/7-01 headset-microphone. The earphone is designed to operate in an ambient peak-noise level of 105 dBA.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, Flat and Seals.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.
PPP-T-60 - Tape: Packaging, Waterproof.
PPP-T-76 - Tape, Packaging, Paper, (For Carton Sealing).

MILITARY

MIL-P-116 - Preservation-Packaging, Methods of.
MIL-C-45662 - Calibration System Requirements.
MIL-H-83511 - Headset - Microphone and Headset - Electrical (Medium Noise Attenuation, Hearing Protective).
MIL-H-83511/7- Headset - Microphone, M83511/7-01.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Air Force Acquisition Logistics Division, Directorate of Electronic Support (AFALD/PTES), Wright-Patterson AFB, OH 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-794 - Parts and Equipment, Procedure for Packaging and Packing of.
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.
MIL-STD-1285 - Marking of Electrical and Electronic Parts.
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC.

S1.12-1967 (R1972) - Laboratory Standard Microphones, Specifications for.
S3.5-1969 (R1973) - Methods for the Calculation of the Articulation Index.
S3.7-1973 - Coupler Calibration of Earphones, Method for (revision and redesignation of Z24.9-1949).

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Qualification. Earphone elements furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3).

3.2 Materials. Materials shall be used which will enable the earphone elements to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.2.1 Insulating and impregnating compounds. All insulating and impregnating compounds applied, including varnishes, waxes and the like, shall be suitable for each particular application. Compounds shall adequately preserve the dielectric strength of the insulation to which it is applied. The compound used shall not (either in the state of its original application or as a result of aging) show any injurious effect upon the insulation it is to protect, and shall not cause corrosion or deterioration of adjacent metal or plastic parts.

3.2.2 Magnetic stability. The materials used in the magnetic circuit shall be processed and assembled so that the earphone element will suffer no degradation in performance over long periods of use or storage.

3.2.3 Cementing compounds. Any cementing compound used in the construction of the earphone element shall show no evidence of loosening nor have other detrimental effects.

3.2.4 Fungus resistant. Unless certification is furnished, materials for fabrication of earphone elements covered by this specification shall conform to requirement 4 of MIL-STD-454. When nutrient materials are used, they shall be tested prior to fabrication of the earphone element.

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3.2.5 Soldering. The materials used for soldering shall conform to requirement 5 of MIL-STD-454.

3.3 Design. Earphone elements shall be designed in accordance with figure 1; however, the configuration is optional.

3.3.1 Diaphragm. The earphone element shall be constructed to have equalization of pressure on both sides of the diaphragm when subjected to changes in ambient pressure as low as 3.4 inches of mercury.

3.3.2 Weight. Unless otherwise specified, the maximum weight of the earphone element shall be 25 grams.

3.3.3 Articulation index. The earphone element and earcup assembly (see 4.3) shall provide a word intelligibility score that could be expected from a distortion-free communications system with a minimum A.I. of 0.50 per MIL-STD-1472 (see 6.4). The A.I. shall be determined in accordance with ANSI S3.5 - 1969.

3.4 Performance characteristics.

3.4.1 Frequency response (see 4.6.2). The frequency response at 1 kilohertz (kHz) shall be 103 ±2 decibels (dB). The response of each earphone element shall not vary from the value at 1 kHz by more than ±5 dB across the frequency response range. Unless otherwise specified, the frequency response range shall be 20 hertz to 6 kHz.

3.4.2 Impedance (see 4.6.3). The impedance at 1 kHz shall be 20 ±2 ohms.

3.4.3 Harmonic distortion (see 4.6.4). The total harmonic distortion shall not exceed 3 percent from 100 Hz to 2 kHz, and it shall not exceed 5 percent from 2 kHz to 6 kHz.

3.4.4 Acoustic quality (see 4.6.5). There shall be no buzzes, rattles, or other spurious sounds that would impair the quality of the audio output signal.

3.4.5 Dielectric withstanding voltage (see 4.6.6). There shall be no arcing or breakdown of the insulation. There shall be no more than 500 microamperes leakage current. (A minimum resistance of 1 megohm shall be maintained).

3.4.6 Stray magnetic field (see 4.6.7). The deflection of the magnetic compass shall be not greater than 5 degrees for any fixed earphone orientation.

3.4.7 Endurance (see 4.6.8). There shall be no audible rattles or other signs of failure, and the frequency response shall not vary more than ±3 dB from the initial reading recorded in 3.4.1.

3.4.8 Moisture resistance (see 4.6.9). The frequency response, harmonic distortion, and dielectric withstanding voltage shall be as specified in 3.4.1, 3.4.3, and 3.4.5, respectively. There shall be no loosening or deformation of parts or other damage.

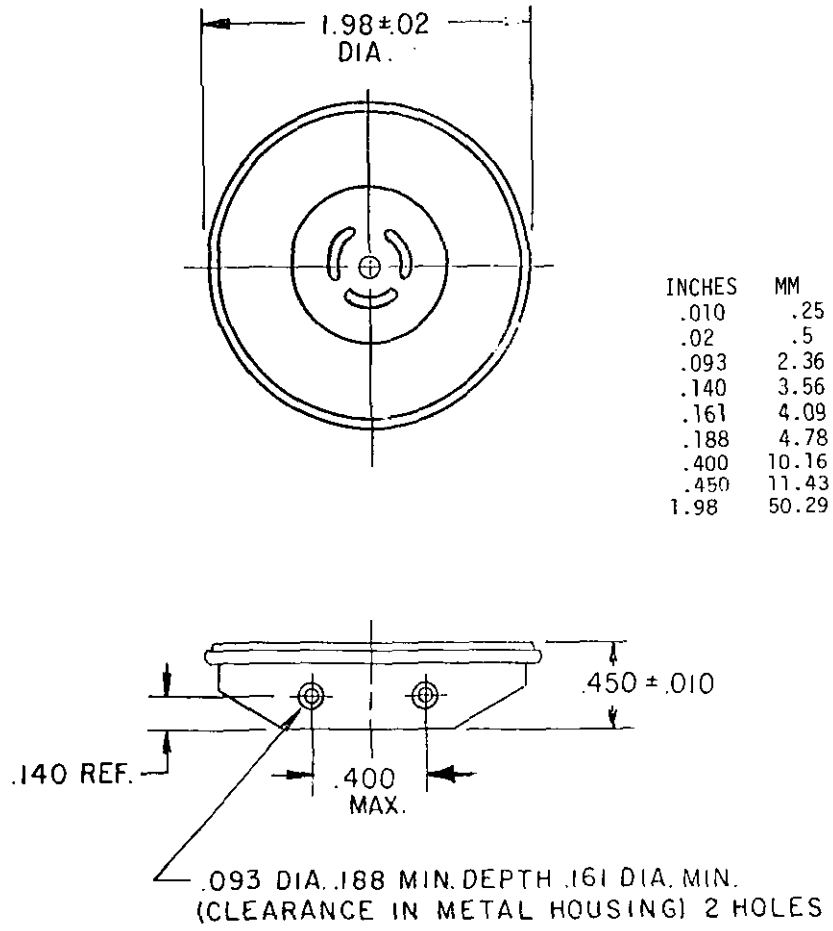
3.4.9 Barometric pressure (reduced) (see 4.6.10). The frequency response shall not vary more than ±3 dB from the initial reading recorded in accordance with 3.4.1.

3.4.10 Thermal shock (see 4.6.11). The frequency response shall not vary more than ±3 dB from the initial reading recorded in accordance with 3.4.1.

3.4.11 Immersion (see 4.6.12). The frequency response shall not vary more than ±3 dB from the initial reading recorded in accordance with 3.4.1.

3.5 Marking. Earphone elements shall be marked in accordance with MIL-STD-1285 with the military part number and the manufacturer's code or name.

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INCHES	MM
.010	.25
.02	.5
.093	2.36
.140	3.56
.161	4.09
.188	4.78
.400	10.16
.450	11.43
1.98	50.29

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Size, number, and location of holes in cap to be determined by manufacturer, subject to acoustical and electrical requirements and limiting dimensions. A single comparatively large hole is not acceptable.
4. The configuration is optional.

FIGURE 1. Earphone element.

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3.6 Workmanship. Earphone elements shall be processed in such a manner as to be uniform in quality and shall be free from loose parts, excess solder, sharp edges, burrs, flaking, scratches, peeling, metal chips, and foreign material. Soldering, impregnation of coils, plating, staking, riveting, and machine screw assemblage shall be neat and thorough. The coil shall be centered properly, void of foreign objects in the air gap, and shall not drag on the frame or magnet.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accurancy of the measuring and test equipment shall be in accordance with MIL-C-45662.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. Qualification inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202, and all electrical-acoustical tests shall be performed with the earphone element mounted in the earcup of the MS3511/7-01 per MIL-H-83511/7.

4.4 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.3) on sample units produced with equipment and procedures normally used in production.

4.4.1 Sample size. Six earphone elements shall be subjected to the qualification inspection.

4.4.2 Inspection routine. The sample shall be subjected to the inspections specified in table I, in the order shown. All sample units shall be subjected to the inspections of group I. The sample shall then be divided equally into two groups of three units each and subjected to the inspection for their particular group.

4.4.3 Failures. One or more failures shall be cause for refusal to grant qualification approval.

4.4.4 Retention of qualification. To retain qualification, the contractor shall forward a report at 12-month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of:

a. A summary of the results of the tests performed for inspection of product for delivery, groups A and B, indicating as a minimum the number of lots that have passed and the number that have failed. The results of tests of all reworked lots shall be identified and accounted for.

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b. The results of tests performed for periodic inspection, group C, including the number and mode of failures. If the test results indicate nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

Failure to submit the report within 30 days after the end of each 12-month period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the contractor shall immediately notify the qualifying activity at any time during the 12-month period that the inspection data indicates failure of the qualified product to meet the requirements of this specification.

In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during two consecutive reporting periods, there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit the earphone elements to testing in accordance with the qualification inspection requirements.

TABLE I. Qualification inspection.

Inspection	Requirement paragraph	Test method paragraph
Group I		
Visual and mechanical examination	3.2, 3.3, 3.5, and 3.6	4.6.1
Frequency response	3.4.1	4.6.2
Impedance	3.4.2	4.6.3
Harmonic distortion	3.4.3	4.6.4
Acoustic quality	3.4.4	4.6.5
Dielectric withstanding voltage	3.4.5	4.6.6
Stray magnetic field	3.4.6	4.6.7
Endurance	3.4.7	4.6.8
Group II		
Moisture resistance	3.4.8	4.6.9
Barometric pressure (reduced)	3.4.9	4.6.10
Group III		
Thermal shock	3.4.10	4.6.11
Immersion	3.4.11	4.6.12

4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspection.

4.5.1.1 Inspection lot. An inspection lot shall consist of all earphone elements of the same part number, produced under essentially the same conditions, and offered for inspection at one time.

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4.5.1.2 Group A inspection. Group A inspection shall consist of the inspections specified in table II, in the order shown.

4.5.1.2.1 Sampling plan.

4.5.1.2.1.1 For subgroup 1. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality level (AQL) shall be as specified in table II. Major and minor defects shall be as defined in MIL-STD-105.

4.5.1.2.1.2 For subgroup 2. The test in subgroup 2 shall be performed on all earphone elements offered for acceptance.

TABLE II. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL (percent defective)	
			Major	Minor
Subgroup 1				
Visual and mechanical examination	3.2, 3.3, 3.5, and 3.6	4.6.1	1.0	4.0
Frequency response	3.4.1	4.6.2	1.0	None
Impedance	3.4.2	4.6.3		
Subgroup 2				
Acoustic quality	3.4.4	4.6.5	Not applicable - 100% inspection	

4.5.1.2.2 Rejected lots (subgroup 1). If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.5.1.3 Group B inspection. Group B inspection shall consist of the inspections specified in table III, in the order shown, and shall be made on sample units which have been subjected to and have passed the group A inspection.

4.5.1.3.1 Sampling plan. The sampling plan shall be in accordance with MIL-STD-105 for special inspection level S-4. The sample size shall be based on the inspection lot size from which the sample was selected for subgroup 1, group A inspection. The AQL shall be 2.5 percent defective.

TABLE III. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph
Harmonic distortion	3.4.3	4.6.4
Dielectric withstanding voltage	3.4.5	4.6.6
Stray magnetic field	3.4.6	4.6.7

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4.5.1.3.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.5.1.3.3 Disposition of sample units. Sample units which have passed all the group B inspection may be delivered on the contract, if the lot is accepted and the sample units are still within specified electrical tolerances.

4.5.2 Periodic inspection. Periodic inspection shall consist of group C. Except where the results of these inspections show noncompliance with the applicable requirements (see 4.5.2.1.4), delivery of products which have passed group C shall not be delayed pending the results of these periodic inspections.

4.5.2.1 Group C inspection. Group C inspection shall consist of the inspections specified in table IV, in the order shown. Group C inspection shall be made on sample units selected from inspection lots which have passed the groups A and B inspections.

4.5.2.1.1 Sampling plan. Four sample units shall be selected from each 2 year's production, or each 6000 units whichever occurs first, after date of notification of qualification. No group C testing is required on lots of less than 100 units.

TABLE IV. Group C inspection.

Inspection	Requirement paragraph	Test method paragraph	Number of sample units to be inspected
Subgroup 1			
Endurance	3.4.7	4.6.8	2
Subgroup 2			
Immersion	3.4.11	4.6.12	2
Barometric pressure (reduced)	3.4.9	4.6.10	
Thermal shock	3.4.10	4.6.11	
Visual and mechanical examination	3.2, 3.3, 3.5, and 3.6	4.6.1	

4.5.2.1.2 Failures. If one or more sample units fail to pass group C inspection, the sample shall be considered to have failed.

4.5.2.1.3 Disposition of sample units. Sample units which have been subjected to group C inspection shall not be delivered on the contract.

4.5.2.1.4 Noncompliance. If a sample fails to pass group C inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspections or the inspection which the original sample failed, at the option of the qualifying activity). Groups A and B inspections may be reinstated: however, final acceptance and shipment shall be withheld until the group C inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

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4.5.3 Inspection of packaging. Except when commercial packaging is specified, the sampling and inspection of the preservation and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements od MIL-STD-129. The inspection of commercial packaging shall be as specified in the contract. (see 6.2).

4.6 Methods of inspection.

4.6.1 Visual and mechanical examination. The earphone element shall be examined to verify that the materials, design, physical dimensions, marking, and workmanship are in accordance with the applicable requirement (see 3.2, 3.3, 3.5, and 3.6).

4.6.2 Frequency response (see 3.4.1). The earphone element shall be mounted on a flat plate coupler of an artificial ear, B&K type 4153 or demonstrated equivalent as shown in the test circuit (see figure 2). Adjust the output from the audio oscillator to 1 milliwatt input to the element at 1,000 Hz as measured by the vacuum tube voltmeter (VTVM No. 1). For qualification inspection, the frequency response shall be measured and recorded in accordance with 4.6.2.1. For quality conformance inspection, the frequency response shall be measured and recorded in accordance with 4.6.2.1 or 4.6.2.2, at the option of the contractor.

4.6.2.1 Automatic recorder. The acoustic output shall be recorded on a direct writing strip graph using an automatic plotter or curve tracer with a minimum writing speed of 10 inches per second and a maximum chart speed of 30 inches per minute. The output shall be recorded throughout the frequency range in the dB value above 0.002 dyne per square centimeter. The writing speed and chart speed shall be noted.

4.6.2.2 Point-to-point. Measure the acoustic output from the calibrated microphone and preamplified unit as indicated by the reading of VTVM No. 2 (see figure 2), and convert this reading to the equivalent dB value above 0.0002 dyne per square centimeter, using the most recent available calibration curve for the test microphone. Measure and record the dB increments of 100 Hz from 100 Hz to 1 kHz and at increments of 250 Hz from 1 kHz to the end of the frequency range. The graph shall show the dB values on the ordinate scale, reading from 80 to 140 dB above 0.0002 dyne per square centimeter, and the Hz value on the abscissa scale, reading from 100 Hz to 10 kHz inclusive.

4.6.3 Impedance (see 3.4.2). Impedance measurements shall be made at 1 kHz with the earphone element mounted on the coupler as shown on figure 2. The 1 kHz voltage applied across the terminals of the earphone element during impedance measurements shall be 0.15 ±0.03 volt root-mean-square (rms). The impedance shall be measured by measuring the voltage across and the current through the earphone element.

4.6.3.1 Voltage method. At the option of the manufacturer, the impedance may be calculated by using a voltmeter. Using a circuit similar to that shown in figure 2, measure the voltage across the 20-ohm resistor and across the earphone element under test. The earphone impedance (z) is equal to:

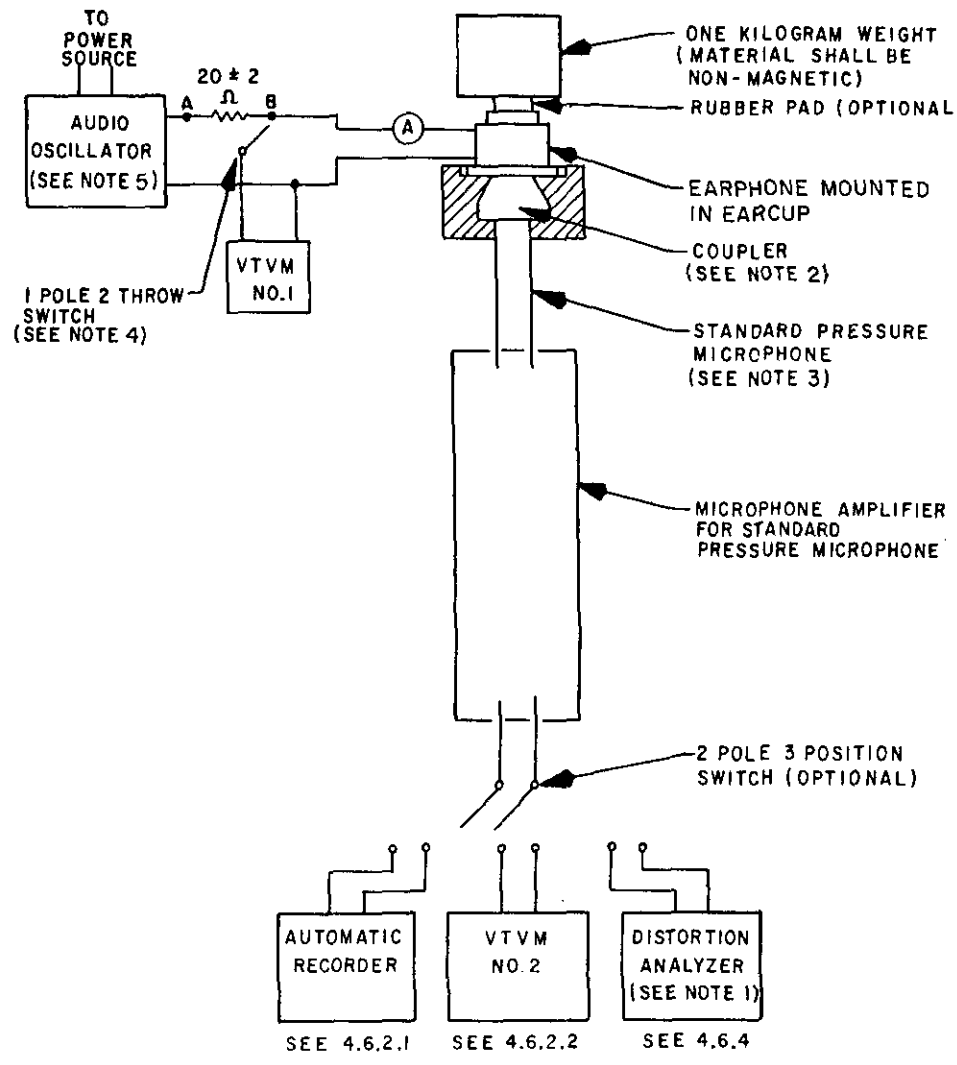
$$20 \times \frac{V_b}{V_a - V_b}$$

V_b is the voltage across the element, and

V_a is the voltage across the element and the resistor.

4.6.4 Harmonic distortion (see 3.4.3). The earphone element shall be tested while connected to the test circuit and mounted on the coupler as shown in figure 2. Distortion shall be measured at 200 Hz and at frequencies between 400 Hz and 6 kHz, as required to determine at what frequencies (in this range) maximum distortion exists.

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NOTES:
1. See 4.6.4.1. for type and requirement for distortion analyzer.
2. Coupler in accordance with flat plate specified see 4.6.2.
3. Standard pressure microphone, in accordance with ANSI S1.12-1967.
4. Use switch position B for paragraph 4.6.3 and switch position A for all other paragraphs requiring electrical tests.
5. Audio oscillator output shall be impedance matched for the circuit.
6. "Off" position on switches is optional.
7. Either voltmeter method or voltmeter-ammeter method is acceptable.

FIGURE 2. Typical test circuit.

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4.6.4.1 Distortion analyzer. The total harmonic distortion shall be determined by a distortion analyzer having the following minimum requirements:

a. Frequency range - Fundamental frequency from 20 Hz to 20 kHz.
b. Frequency calibration - ±2 percent from 20 to 20,000 Hz.
c. Harmonic measurement - ±3 percent of full scale value for distortion levels as low as 0.5 percent.

4.6.5 Acoustic quality (see 3.4.4). The 0.15 ±0.03 volt rms shall be applied continously to the terminals of the earphone element. The frequency shall be varied continously from 20 Hz to 7 kHz and back to 20 Hz. The acoustic output shall be checked for buzzing, rattling, and other spurious sounds.

4.6.6 Dielectric withstanding voltage (see 3.4.5). The earphone element shall be tested in accordance with method 301 of MIL-STD-202. The following details and exception shall apply:

a. Magnitude of test voltage - 500 volts.
b. Nature of potential - 60 Hz ac, or direct current
c. Points of application - Between one of the terminals and the case.
d. After the application, the current leakage and resistance.

4.6.7 Stray magnetic field (see 3.4.6). Stray magnetic field measurements shall be made in a shielded room or in an area free from appreciable magnetic disturbances. The earphone element shall be rotated on its axis at various points of a 12-inch radius sphere surrounding a Kueffel and Esser type No. 80-0090 compass, or equal.

4.6.8 Endurance (see 3.4.7). Audio power of 300 milliwatts at 1 kHz shall be applied continuously to each earphone element under test for 8 hours, minimum. During the last hour of the test, the earphone element shall be checked for audible rattles or other signs of failure. At the end of the 8 hours, the frequency response shall be measured as specified in 4.6.2.

4.6.9 Moisture resistance (see 3.4.8). The earphone element shall be tested in accordance with method 106 of MIL-STD-202. The following details and exception shall apply:

a. Initial measurements - Not applicable.
b. Mounting - Any convenient mounting with the front face of the earphone element exposed and parallel with the vertical plane.
c. Loading voltage - Not applicable.
d. Final measurements - At the completion of the tenth cycle and following a 24-hour period at 25° ±5°C and 50 ±5 percent relative humidity, the earphone element shall be tested for frequency response, harmonic distortion, and dielectric withstanding voltage in accordance with 4.6.2, 4.6.4, and 4.6.6, respectively, and shall then be examined for loose or deformed parts and other damage.

4.6.10 Barometric pressure (reduced) (see 3.4.9). The earphone element shall be tested in accordance with method 105 of MIL-STD-202. The following details shall apply:

a. Method of mounting - Not applicable.
b. Test condition letter - B.
c. Tests during subjection to reduced pressure - None required.
d. Tests after subjection to reduced pressure - Frequency response in accordance with 4.6.2.
e. Exposure time prior to measurements - 10 minutes.

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4.6.11 Thermal shock (see 3.4.10). The earphone element shall be tested in accordance with method 107 of MIL-STD-202. The following details shall apply:

a. Test condition letter - A.
b. Measurement after cycling - Frequency response in accordance with 4.6.2.

4.6.12 Immersion (see 3.4.11). The earphone element shall be submerged in a 3 ft. head of water for 1 minute. The element shall be removed from the water, wiped dry with a cloth, and the frequency response shall be measured as specified in 4.6.2.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, C or commercial, or as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Earphone elements shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Earphone elements shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. Preservatives shall not be used.

5.1.1.4 Unit packs. Each earphone element shall be individually unit packed in accordance with MIL-P-116, submethod IA-8 insuring compliance with the applicable requirements of that specification.

5.1.1.5 Intermediate packs. Earphone elements, packaged as specified in 5.1.1.4, shall be placed in intermediate containers conforming to PPP-B-566 or PPP-B-676. Intermediate containers shall be uniform in size, shape and quantities, shall be of minimum tare and cube and shall contain multiples of five unit packs, not to exceed 100 unit packs. No intermediate packs are required when the total quantity shipped to a single destination is less than 100 unit packs.

5.1.2 Level B. The level B preservation for earphone elements shall be identical to that specified for level A except that submethod IC-1 or IC-3 of MIL-P-116 shall be used in lieu of submethod IA-8.

5.1.3 Level C. The level C preservation for earphone elements shall conform to the MIL-STD-794 requirements for this level.

5.1.4 Commercial. The commercial preservation of earphone elements shall be in accordance with the requirements of MIL-STD-1188.

5.2 Packing. Packing shall be level A, B, C or commercial, or as specified (see 6.2).

5.2.1 Level A. The packaged earphone elements shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. In lieu of the closure and waterproofing requirement in the appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joints with tape, two inches minimum width, conforming to PPP-T-60, class 1 or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636 using nonmetallic or tape banding only.

5.2.2 Level B. The packaged earphone elements shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements. Closures shall be in accordance with the appendix thereto.

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5.2.3 Level C. The level C packing for earphone elements shall conform to the MIL-STD-794 requirements for this level.

5.2.4 Commercial. The preserved earphone elements shall be packed in accordance with the requirements of MIL-STD-1188.

5.3 Marking.

5.3.1 Levels A, B and C. In addition to any special marking required by the contract (see 6.2), each unit pack, intermediate and exterior container shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial marking shall be in accordance with the requirements of MIL-STD-1188.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2 and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.3.

5.4.3 Army procurements.

5.4.3.1 Level A intermediate packs. In addition to the requirements of 5.1.1.5, all intermediate containers shall be either weather (or water) resistant or overwrapped with waterproof barrier materials.

5.4.3.2 Level A and Level B packing. For level A packing the fiberboard containers shall not be banded but shall be placed in a close fitting box conforming to PPP-B-601, overseas type; PPP-B-621, class 2, style 4 or PPP-B-585, class 3, style 2 or 3. Closure and strapping shall be in accordance with applicable container specification except that metal strapping shall conform to QQ-S-781, type I, finish A. When the gross weight exceeds 200 pounds or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids (laid flat) shall be applied in accordance with the requirements of the container specification. If not described in the container specification, the skids shall be applied in a manner which will adequately support the item and facilitate the use of material handling equipment. For level B packing, fiberboard boxes shall be weather resistant as specified in level A and the containers shall be banded (see 5.2.1 and 5.2.2).

6. NOTES

6.1 Intended use. Earphone elements covered by this specification are mounted on the inside of the earcup of headsets which have an extended range and which are used in medium ambient noise conditions; e.g. headset-microphone M83511/7-01.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Inspection of commercial packaging (see 4.5.3).
c. Levels of preservation and packing required (see 5.1 and 5.2).
d. Special marking, if required (see 5.3.1).

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts for the products covered by this specification. The activity responsible for the Qualified Products List is the Air Force Acquisition Logistics Division, Electronic Support Division (AFALD/PTES), Wright-Patterson AF Base, OH 45433; however, information pertaining to qualification of products may be obtained from the Defense Electronics Supply Center (DESC-EQ), Dayton, OH 45444. (see 4.5.2)

6.4 Articulation index correlation. The A.I. must be correlated with the real noise environment and the flat plate. See your contracting officer.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy -
Air Force - 11,99
DLA - ES

User activities:
Army - MD
Navy - OS, AS
Air Force

Preparing activity:
Air Force - 85

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